Module: Corporate Finance & Equity Valuation

School of Economics

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Leasing

Learning Objectives

- Understand the types of leases and how each type provides the right to use the asset.
- Identify the differences between operating and financial leases from a financial management and financial reporting perspective.
- Describe a direct lease, sale and lease-back and a leveraged lease.
- Identify the advantages of leasing.
- Be able to calculate the net present cost (NPC) of the leasing option.
- Be able to calculate the NPC of the borrow and buy option.
- Be able to: (a) calculate the net advantage of leasing (NAL) and (b) advise whether a company should lease or borrow and buy.

Leasing – An Introduction

- A lease is a contract providing a right to the use of an asset, legally owned by the lessor, in exchange for a specified rental paid by the lessee.
- **Leasing is a source of finance.**
- Lease rentals (normally payable at the beginning of each month) are included in the gross taxable income of the lessor. The lessee deducts the lease rentals when calculating taxable income.
- At the end of the lease term, the lessee would have the following options:
  - Renewing the lease,
  - Acquiring the asset, or
  - Returning the asset to the lessor).
- From the accounting perspective, IAS 17 identifies two types of leases: **a finance lease and an operating lease**.

Types of Leases

- **<u>There are two broad types of Leases.</u>**

<u>1. An operating lease</u> is a lease that does not transfer the risks and rewards of ownership.

- It is an arrangement that provides an alterative means of obtaining the use of an asset without purchasing it.
- Operating lease is normally characterised as short term in nature in relation to the useful life of the asset being leased.
- Due to the short term nature of operating lease, the lessor frequently intends to either lease the asset again or sell it at the termination of the lease.
- The Lessee would not have the intention to acquire the use of an asset on a permanent basis.
- The lessor assumes the responsibilities for the maintenance and insurance of the asset.
- Usually operating leases make provisions for cancellation and a required notice period for cancellation would be stipulated.
- Cancellation could also be at the option of the Lessor or Lessee.
- Operating leases are not considered as an alternative source of finance because the use of the asset acquired this way would not extend to the usefulness of the asset.

Types of Leases

2. A financial lease is a lease that transfers substantially all the risks and rewards associated with ownership of an asset from the lessor to the lessee.

- Characteristics of a financial lease
  - Provides the lessor with reimbursement for the cost of the leased asset plus interest
  - Often the lease agreement allows the Lessee to the option to buy the asset at the termination of the lease.
  - Maintenance and insurance of the leased asset are the responsibility of the Lessee.
  - Financial leases are commonly used to finance the acquisition of motor vehicles, equipment and plant.
  - Therefore, the financial characteristics of a financial lease make it the equivalent of debt financing from the point of view of both the Lessor and the Lessee.
  - Lease payments are enough for the Lessor to recoup both the original cost of the asset, which could be equivalent to the principal amount if the money was borrowed and earn a return on the financing provided, which equates to the to interest.
  - Normally, financial leases are structured in a manner similar to instalment loans.
  - In the event of cancellation of the lease, the Lessee will give a notice of intention to settle the account or forfeit some of the interest that has been charged.
  - To cancel, the Lessee will have to pay the Lessor the account settlement figure which consists of the outstanding capital balance and in some cases with an interest penalty.

Structuring leases

- The different methods used to acquire assets under leases can take the following form: (focus on the 3 most important types).
  - A direct lease
    - In a direct lease the lessor acquires the asset from the manufacturer in order to lease it to the Lessee. Sometimes the manufacturer acts as a Lessor.
  - A sale and lease-back
    - A company sells an asset it owns under an agreement to lease it back.
  - A Leveraged lease
    - The Lessor does not provide all the funding required to purchase the asset that is to be leased. The balance of the funds is obtained from a financial institution. The lease payments made by the Lessee are then used by the Lessor to service the loan. Any excess of the lease payments (part of the lease payments that remains after servicing the loan) is kept or retained by the Lessor.
    - An attraction for the leveraged lease, from the Lessor's point of view, is that there is a potential profit provided by the use of financial leverage and the tax deductions.
    - The risk to the Lessor is not only limited to his/her investment because if the Lessee defaults, the Lessor will have a financial obligation to the financial institution that has lent the funds.
    - Leveraged financing is used to fund large projects.

Criteria for a Finance Lease

- If any ONE of the following is met, then the lease is considered to be a finance lease:
  - The lease transfers ownership of the asset to the lessee by the end of the lease term.
  - The lessee has the option to purchase the asset at a price expected to be significantly lower than its fair value at the date the option becomes exercisable.
  - The lease term is for the major part of the economic life of the asset (e.g. > 75% of the economic life).
  - The PV of the minimum lease payments amounts to at least substantially all of the fair value of the leased asset.
  - The leased assets are of a specialised nature and only the lessee can use them without major modifications.
  - Other indicators – (a) if the lessee can cancel the lease and losses are borne by the lessee; (b) gains and losses in the fair value fall on the lessee; (c) the lessee has the ability to continue the lease for a secondary period at a rent which is substantially lower than the market rent.

Lease Accounting

- Finance leases are essentially treated as debt financing:
  - Present value of lease payments must be included on the balance sheet as a "finance lease liability".
  - The same amount (or cost of the asset) is recognized in the books of the lessee and reported on the balance sheet.
- Operating leases are still "off-balance sheet"
  - "off balance sheet because the Lessee does not own the asset, thus the leased asset and liability reflected in the lease contract may not be reflected in the balance sheet, hence the assets, liabilities and the debt-equity ratio of the lessee are understated.
- The proposed IFRS on leases will require that ALL leases (with a lease-term greater than one year) be treated as finance leases. The lessee will recognize:
  - A "right-to-use" asset on the Balance sheet; and
  - A liability being the "lease payments". Both being the Present value of future lease payments.

The Benefits of Leasing

- Rapid change in technology- advantage is that the term of the lease can be considerably shorter than the life of the leased asset. Cancellation clause on long term leases can be exercised at the option of the Lessee. The above allows the lessee to avoid the risk of an asset becoming obsolete or redundant prior to its originally estimated life.
- Tax advantages- lower combined tax paid is by the Lessor and Lessee relative to the tax paid if the lessee were to buy the asset.
    - e.g. if a firm has assessed tax loss, or does not expect sufficient income to utilize the depreciation deduction and interest tax shield generated by owning the plant, then by leasing the plant from a profitable Lessor, the Lessee can transfer these tax shields to the Lessor. The Lessor would then lower the required lease payments, and as a result, the savings would be shared between the lessee and the lessor.
- Obtaining 100% debt financing- advantage to the Lessee is to acquire the use of the asset without providing the cash investment as an alternative to paying the deposit and purchasing the asset on an instalment sale.

The Benefits of Leasing (continued)

- Operating flexibility – allows a company to react quickly to changing market conditions. E.g. an Airline- if a certain route is not profitable and other routes are,- if a firm requires a different type of aircraft that uses a profitable route, it would be able to maximize operational flexibility by having operating leases rather than owning an aircraft.
- Reduction in operating leverage- the company may be able to enter into an agreement that requires the company to pay an amount per use rather than per period. This enables the company to change a fixed cost to a variable cost and lessors are able to charge a greater fee. The lessors may also be able to diversify risk as he/she will be able to lease the assets to many customers.
- Coping with uncertain demand- if the company has an uncertain demand for its products it is able to reduce exposure to fixed costs and long term commitments by undertaking a lease with a cancelation clause. If demand increases substantially, leasing enables the company to increase supply by leasing the asset.
- Specialization effects on maintenance, residual values and purchase costs.
- Standardization of contracts- leasing is easier to arrange from an administrative view, because it is subject to standardized contracts.

Benefits of Leasing (continued)

- **Fewer restrictions-** leasing may require fewer restrictions than loans. Lessor may require less financial and credit information. (lessor's security is the asset and he/she understands the value and risk of the leased asset).
- **Off-statement of financial position financing-** In terms of IAS17, if a company is able to avoid the requirements of the standard relating to finance leases, then the lease will be classified as an operating lease and will not be reflected in the balance sheet, hence, resulting in a perception of a lower financial leverage. However, not that:in the past investors may have tried to have "of balance sheet" accounting by taking on leases; this should not happen anymore because of tighter accounting rules.
- **Avoidance of capital expenditure controls and budgetary constraints-** companies may put stringent requirements on capital budgeting than on leasing decisions. Budgetary constraints will encourage leasing of assets by companies.

Evaluating the Lease or Buy Decision

- Normally, the investing decision is evaluated first. This entails using WACC to discount the following investment cash flows:
  - Purchase price. **No discounting required.**
  - Tax shield on wear and tear allowances. $(i \times T\%)$
  - Revenue and expenses.
  - Income tax.
  - Residual value.
- **After the investing decision (e.g. positive NPV), the company needs to make the financing decision: That is: Is it better-off borrowing and buying the asset or leasing?**
- Remember that leasing is a financing decision.

Evaluating the Lease or Buy Decision

The following steps are needed to determine whether to borrow and buy or lease:

- **First**, **calculate the net present cost (NPC) of the leasing option.** The following leasing cash flows are discounted using the **after tax cost of debt** (a lease has the same risk as a company's debt)**:**
  - Lease payments. **Cash outflow**
  - Tax shield on lease payments. **Cash inflow**
  - Maintenance and insurance (if applicable). **Cash outflow**
  - Tax shield on maintenance and insurance (if applicable). **Cash inflow**
  - **Second**, **calculate the NPC of the borrow and buy option**. The following borrow and buy cash flows are also discounted using the **after tax cost of debt:**
  - **Loan repayments and tax shield on interest expense. The PV of these cash flows is normally the same as the original capital sum. Refer to Lecture Example 1.**
  - Tax shield on wear and tear (tax allowances).
  - Residual value of the assets and any related recoupment tax.
  - Maintenance and insurance expense and related tax shield.
- **Finally**, **calculate the net advantage of leasing (NAL): NPC of borrow and buy minus the NPC of leasing. If the NAL is positive (> 0) the firm should lease, if negative (< 0), then borrow and buy. Choose the cheaper option.**

Asset Specific Finance

- Where finance is asset specific, there will be a ***link between*** the investment decision and the financing decision.
- **The evaluation is as follows:**
  - Calculate the NPV. (investment decision)
  - Calculate the NPC of each alternative. (financing decision)
  - Calculate the ***net advantage*** of the asset specific finance by deducting the NPC of asset specific finance from the NPC of borrowing.
  - Add the net advantage of the asset specific finance to the original NPV.
  - If the total is positive, accept the project. Otherwise, reject the project.

Example 1: Leedy Manufacturing

- WACC 15% only used for NPV calculation
- After tax cost of debt 9% different ∴ amortize
- Specific loan for this project 11%
- When the rate of specific debt finance and your after tax cost of debt are different we have to amortize the loan to get the interest.
- When they are the same – simply put cost of loan in year 0
- In the case of Leedy manufacturing, they are not the same, so we need to amortise the loan

Example 1: Leedy manufacturing

Things to note:

This example involves making two decisions:

1. The investment decision- capital budgeting decision- using the NPV method for deciding on whether to accept or reject the investment.
2. The financing decision- which requires us to make a decision whether to lease or to borrow and buy the asset.

Note that: the investment decision starts before the financing decision can be made. If the investment is rejected then there is no financing decision to be made and vice versa.

Example 1: Leedy (workings)

Calculation of Recoupment/scrapping allowance

w1

Working 1

Tax val < resid Tax val > resid

Old Machine (calculation of recoupment/scrapping allowance)

Given in notes – Tax Value = 0

Residual value = R18

Recoupment value (old machine) = Residual value – Tax Value

Recoupment (old machine) = 18 – 0

Recoupment value (old Machine) = 18

w2

Working 2

New Machine (calculation of recoupment/scrapping allowance)

Sec 12C depreciation $33^{1/3}$ % per annum over 3 years

Depreciation per year sec 12C = 480.00/3 = 160.00

Residual value = R20.00

At cost	480.00
Less. Accumulated depreciation	480.00
Tax Value	0.00

Recoupment (new Machine) = 20.00 – 0 which is = 20.00

w3

Leedy working 3 (Taxation schedule)

Year	1	2	3
Incremental cash flows	240.00	245.00	250.00
Less. Maintenance costs	(7.00)	(7.00)	(7.00)
Net Cash flows	233.00	238.00	243.00
Recoupment old machine	18.00		
Recoupment new machine			20.00
Less. Depreciation - section 12 C	(160.00)	(160.00)	(160.00)
Taxable income	91.00	78.00	103.00
Taxation % 28%	25.5	21.8	28.8

Leedy (NPV schedule)

Year	0	1	2	3
Initial investment in machine	(480.00)			
Residual value (old machine)	18			
Incremental cash flows		240.00	245.00	250.00
Less. Maintenance costs		(7.00)	(7.00)	(7.00)
Net Cash flows		233.00	238.00	243.00
Residual value (new machine)				20.00
Taxable income		(25.5)	(21.8)	(28.8)
Net Cash flows		207.5	216.2	234.2
Discount factors @ 15% WACC $(1+15\%)^n$		0.8696	0.7561	0.6575
PVs of cash flows $\frac{FV}{(1+15\%)^n}$		180,45	163,45	153,96
Sum of Pvs of cash flows	497.86			
NPV	**35,86**			

Leedy NPC Leasing

R187.0 x 28% = R52.4 Remember tax is calculated at the end of the year

Year		0	1	2	3
Lease payment		-187.0	-187.0	-187.0	
Tax shield on lease payments	(LP x T %)		52.4	52.4	52.4
Net cash flow		-187.0	-134.6	-134.6	52.4
PVIF 9%		1.0000	0.9174	0.8417	0.7722
Present value of net cash flow		-187.0	-123.5	-113.3	40.4
Net present cost of lease		-383.4			

The PV rate is always after tax cost of debt not WACC

Leedy NPC Borrowing

Loan Amortisation schedule

Year	Instalment/ periodic Payment	i Interest portion tax shield is on i portion only	Capital repayment portion	Capital Balance/ outstanding
0				480.0
1	196.4	= 52.8	+ 143.6	336.4
2	196.4	37.0	159.4	177.0
3	196.4	19.5	177.0	0.0

Note that the instalment/ periodic payment is made up of two parts; that is the interest portion and the portion of the principle/loan repayment. However, the tax shield is on the interest portion only because interest portion is recorded in the Income Statement as an expense, and hence, as the effect of reducing taxable income, which in turn results in a tax saving or benefit to the firm.

Leedy NPC Borrowing

R480 cost / 3 = R160 per annum x 28% = R44.8

R7 cost x 28% = R2

Year		0	1	2	3
Loan repayment			-196.4	-196.4	-196.4
Tax shield on Interest	(i x T%)		14.8	10.4	5.5
Tax shield on S12C allowance			44.8	44.8	44.8
Maintenance charge			-7.0	-7.0	-7.0
Tax shield on maintenance charge	(MC x T%)		2.0	2.0	2.0
Residual value					20.0
Tax on recoupment					-5.6
Net cash flow			-141.9	-146.3	-136.8
PVIF 9%		1.0000	0.9174	0.8417	0.7722
Present value of net cash flow			-130.16	-123.14	-105.64
Net present cost of borrowing		-358.95			

Year	Payment	i Interest portion	Capital portion	Capital Balance
0				480.0
1	196.4	52.8	143.6	336.4
2	196.4	37.0	159.4	177.0
3	196.4	19.5	177.0	0.0

(i x T%)

R52.8 x 28% = R14.8

Example 2

Trinity Trucking (Pty) Ltd

- WACC 11% only used for NPV calculation
- Before tax cost of debt K_d 10% same ∴ no amortization
- Specific loan for this project 10%
- When the rate of specific debt finance and your cost of debt are different we have to amortize the loan to get the interest.
- When they are the same – simply put the cost of loan in year 0
- In the case of Trinity Trucking (Pty) Ltd, they are the same, so **<u>we simply put the cost of the loan under year 0</u>**

Trinity Trucking (Pty) Ltd

Loan amortisation schedule

1. Calculation of loan instalment (I):

NPV = 1.8m = PVA(n=5, i=10%, PMT = ?)

Thus PMT = 474 835.46

$$PVA = PMT\left[\frac{\{1-(1+\frac{i}{m})^{-nm}\}}{\frac{i}{m}}\right] \quad \text{thus } PMT = \frac{PVA}{\frac{\{1-(1+\frac{i}{m})^{-nm}\}}{\frac{i}{m}}}$$

$$PMT = \frac{1\,800\,000}{\frac{\{1-(1+0{,}1)^{-5}\}}{0{,}1}} \qquad \text{thus } PMT = 474\,835.46$$

Trinity Trucking (Pty) Ltd

Loan amortization table for the borrow & purchase option:

Year	Opening balance	Instalment/ periodic Payment	Interest portion	Capital repayment portion	Capital balance/ outstanding
1	1 800 000.00	474 835.46	180 000.00	294 835.46	1 505 164.54
2	1 505 164.54	474 835.46	150 516.45	324 319.01	1 180 845.53
3	1 180 845.53	474 835.46	118 084.55	356 750.91	824 094.63
4	824 094.63	474 835.46	82 409.46	392 426.00	431 668.63
5	431 668.63	474 835.46	43 166.86	431 668.60	0.03

Net present cost (NPC) of borrowing and buy option:

NPC borrow and buy option

	0	1	2	3	4	5
Loan repayment		-474835,47	-474835,47	-474835,47	-474835,47	-474835,47
Interest tax shield (i x T%)		50400	42144,607	33063,6746	23074,6491	12086,7209
Maintenance cost		-60000	-60000	-60000	-60000	-60000
Maintenance tax shield (28%)		16800	16800	16800	16800	16800
Wear and tear allowance tax shield 33.33% x R1.8m x 28%		168000	168000	168000		
Residual value						50000
Recoupment tax on residual value 28% x (proceeds - tax value)						-14000
Total cash flows		-299634,47	-307888,86	-316968,79	-494956,82	-469943,74
PVIF at 7.2% [10% x (1-28%)]		0,9328	0,8702	0,8117	0,7572	0,7064
NPC (net present cost)		-279509,76	-267919,55	-257295,47	-374790,15	-331949,44
Total Net Present Cost (NPC)	-1511464					

Alternative calculation

When the rate of specific debt finance and your cost of debt are the same – simply put the cost of loan in year 0 (see below) For this question they are both 10%

	0	1	2	3	4	5
Loan repayment	-1800000					
Maintenance cost		-60000	-60000	-60000	-60000	-60000
Maintenance tax shield (28%)		16800	16800	16800	16800	16800
Wear and tear allowance tax shield 33.33% x R1.8m x 28%		168000	168000	168000		
Residual value						50000
Recoupment tax on residual value 28% x (proceeds - tax value)						-14000
Total cash flows	-1800000	124801	124802	124803	-43196	-7195
PVIF at 7.2% [10% x (1-28%)]	1	0,9328	0,8702	0,8117	0,7572	0,7064
NPC (net present cost)	-1800000	116418,843	108600,537	101307,283	-32708,783	-5082,2599
Total Net Present Cost (NPC)	-1511464					

Net present cost (NPC) of the leasing option:

	0	1	2	3	4	5
Lease payments	-400 000.00	-400 000.00	-400 000.00	-400 000.00	-400 000.00	
Lease payment tax shield		112 000.00	112 000.00	112 000.00	112 000.00	112 000.00
Maintenance cost		-60 000.00	-60 000.00	-60 000.00	-60 000.00	-60 000.00
Maintenance tax shield		16 800.00	16 800.00	16 800.00	16 800.00	16 800.00
Total cash flows	**-400 000.00**	**-331 200.00**	**-331 200.00**	**-331 200.00**	**-331 200.00**	**68 800.00**
PVIF at 7.2% [10% x (1-28%)]	1.00	0.9328	0.8702	0.8117	0.7572	0.7064
NPC (net present cost)	-400 000.00	-308 955.22	-288 204.50	-268 847.48	-250 790.56	48 597.57
Total NPC: Leasing option	**-1 468 200**					

Deciding on which option to choose

NPC of the borrow and buy option = 1 511 464

NPC of the leasing option = 1 468 200

There is a net advantage of leasing of R43 264.

The Leasing option has a lower Net Present Cost (NPC) thus, it is preferred since it's the cheaper option.